

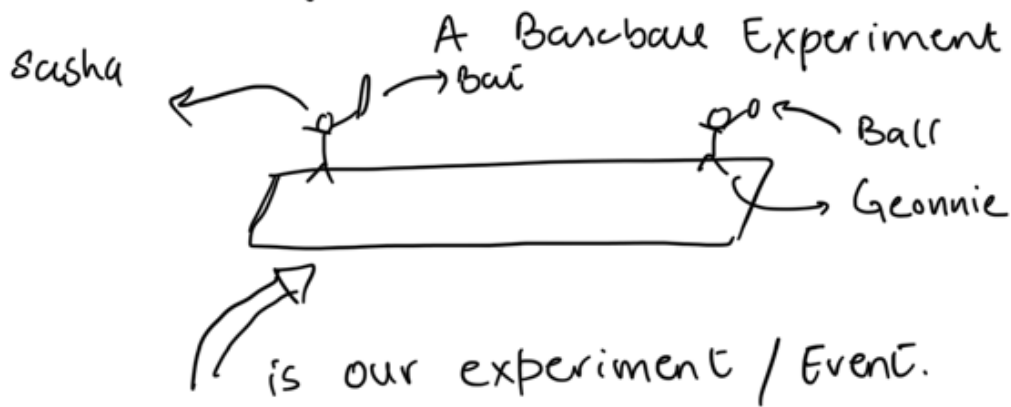
Probability

Event ✓

Outcomes ✓

Sample space ✓

Probability of an Outcome ✓



That this event can lead to 4 things.

1. Sasha misses the ball } A
 2. Sasha hits the ball } B
 3. Sasha hits a homerun } C
 4. Sasha gets hit by the ball } D
- } these are the outcomes of our Event.

Then we let this experiment happen
a collected data for the outcomes.

This experiment took place for 100

Образец instances. $\Rightarrow$ the following data.
Космос

sample space { 30 times sasha missed } outcome A
40 times sasha hit the ball
20 times sasha hit a homerun
10 times sasha got hit by the ball

Because we get this by
sampling our event

Probability of = Number of times an
Any Outcome outcome happens
Size of sample
space.

Probability of Sasha = 30/100
missing the ball

Represent this by
writing

$P(\text{Sasha missing the Ball})$

$P(A) = 30/100 \leftarrow 0.3$

Axioms of Probability.

→ Axioms are things/Rules in Mathematics that you "accept" to be true.

[Even without proof].

$$1+1=2$$

→ Real Analysis

Axioms in Probability

1. $0 \leq P(A) \leq 1$

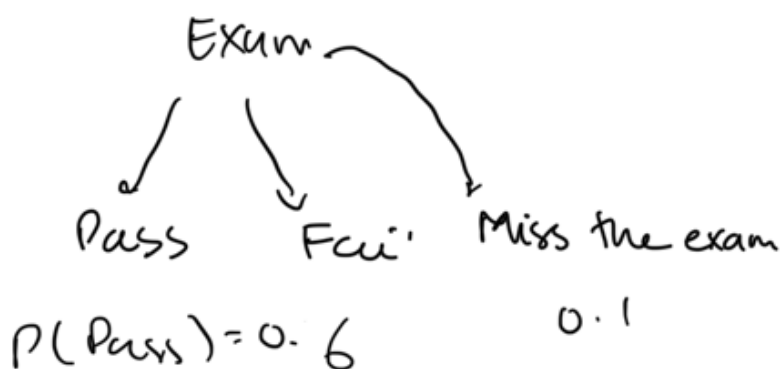
↗ Any outcome to any event

2. The sum of the probabilities of all the outcomes related to an event is equal to 1.

$$P(A) + P(B) + P(C) + P(D) = 1$$

$$0.3 + 0.4 + 0.2 + 0.1 = 1$$

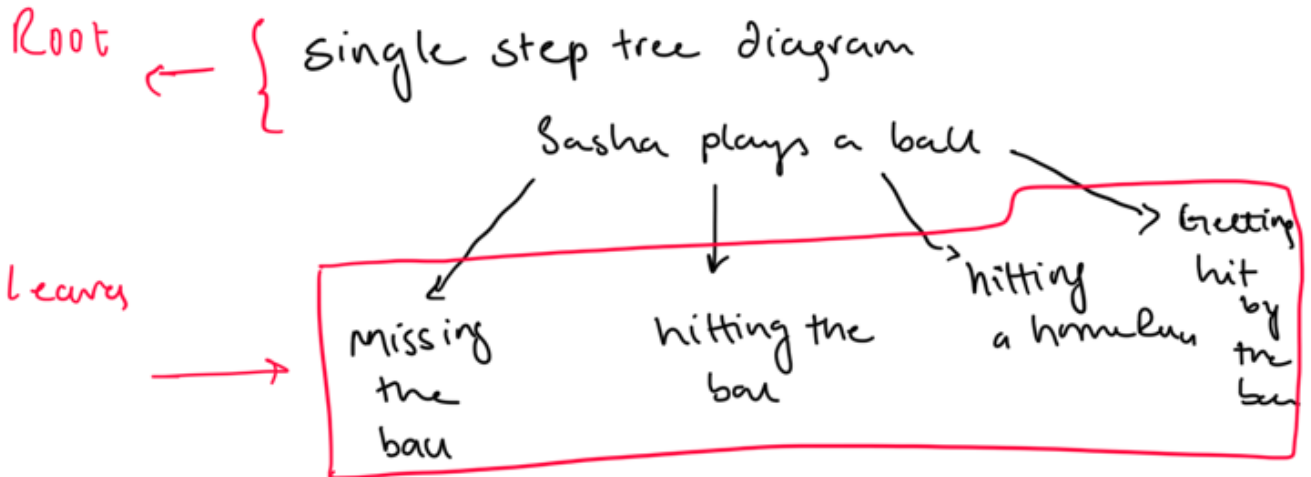
Tree diagram.



$$P(\text{Fail}) = 0.3$$

$$P(\text{Miss}) = 0.1$$

$$P(\text{Pass or Fail or Miss}) = 0.6 + 0.3 + 0.1 = 1$$



Let's say I have one win

coin → { Heads, Tails }
H, T

here I am only interested in a single coin toss.

A tree for this

Sample space / total Outcomes.

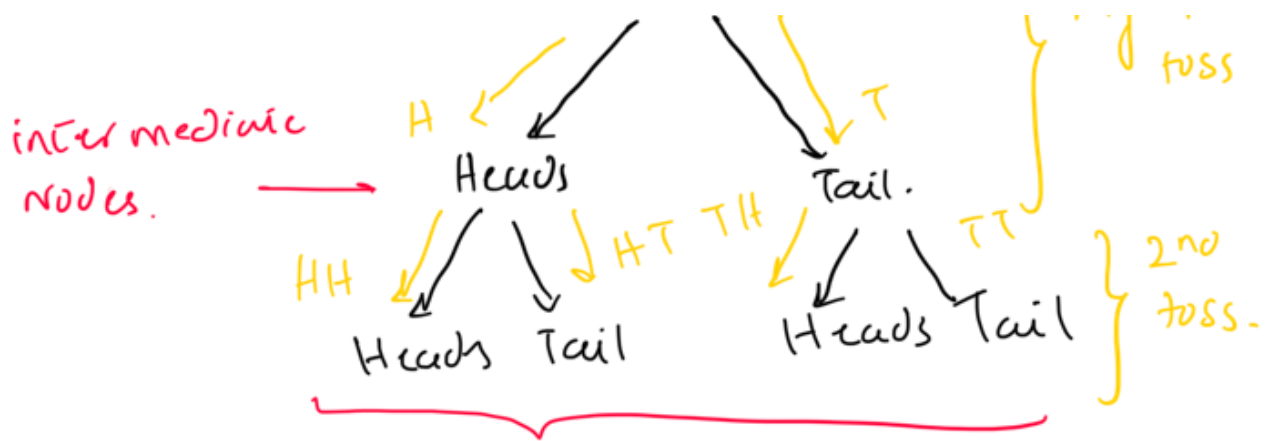
toss the win
↓
Heads
↓
tails.

$$P(H) = \frac{1}{2}, P(T) = \frac{1}{2}$$

Now what if I saw, that I am interested in 2 consecutive coin tosses?

Root — toss the win

1 My first



Heads (H)
Tails (T)

Outcomes / Sample Space

this matters a lot.

$= \{ HH, HT, TH, TT \}$

$$P(HH) = \frac{1}{4}$$

$$P(TH) = \frac{1}{4}$$

$$P(HT) = \frac{1}{4}$$

$$P(TT) = \frac{1}{4}$$

$$P(HH) + P(HT) + P(TH) + P(TT) = 1$$

i had 3 people & i
wanted to make a group of
two people.

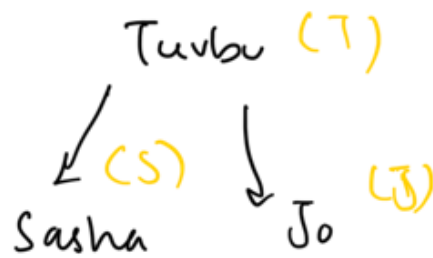
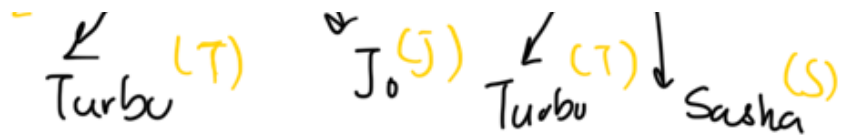
Sasha, Turbo, Jo

i want to choose two
people from here

Sasha (S)

Jo (J)

$\{ST, SJ, \textcircled{JT}, JS, TS, \textcircled{TJ}\}$



is there a

difference between TJ & JT. **No!**

because they would
be on the same team.

$$P(\text{Joe \& Turbo on the same team}) = \frac{2}{6}$$

why $\Rightarrow$ 2 cases JT, TJ

Because order
doesn't matter.

Another example of drawing tree
diagrams.

You toss a dice two times.



Your sample space will have how many

elements? $\{6 \times 6\} \leftarrow$ second toss
 $\leftarrow$ first toss $= 36$ elements

that means any particular outcome

like $(1, 3)$ has probability

$\nearrow$ First toss $\nwarrow$ Second toss

$$\frac{1}{36}$$

if an event or series of m events $m=4$ (in our Example).

take place where each event had

$a_1, a_2, a_3 \dots a_m$ outcomes then the

total # of ways the entire events

can happen together is $a_1 \times a_2 \times a_3 \dots a_m$,

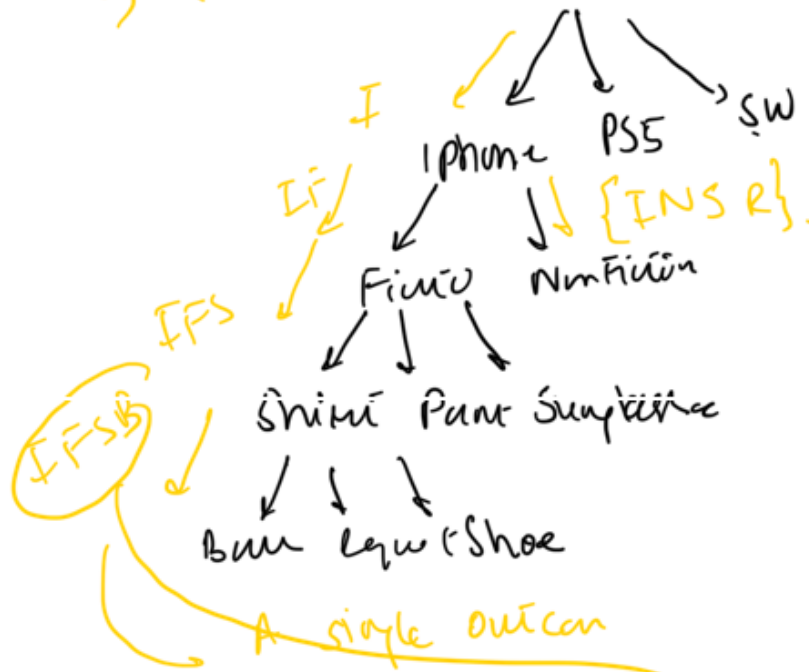
provided the events don't influence

each other.

Let's say you go out shopping
 & you will buy something in the
 following genre.

1. Tech item. $\{ \text{iphone, PS5, Smart watch} \}$. $a_1=3$
2. Book $\{ \text{Fiction, Non-Fiction} \}$. $a_2=2$

3. Clothing item {Shirt, Pant, Sunglasses}. $\{n_3 = 3\}$
 4. Sports item {Ball, Racket, Shoes}. $\{n_4 = 3\}$
 4 Events



total outcome

$$= n_1 \times n_2 \times n_3 \times n_4$$

$$= 3 \times 2 \times 3 \times 3$$

$$= 6 \times 9$$

$$= 54$$

$$\text{Probability of } \{IFS\} = \frac{1}{54}$$

Problem - 1

if the probability of winning a competition is 0.45, what is the probability of losing that competition?

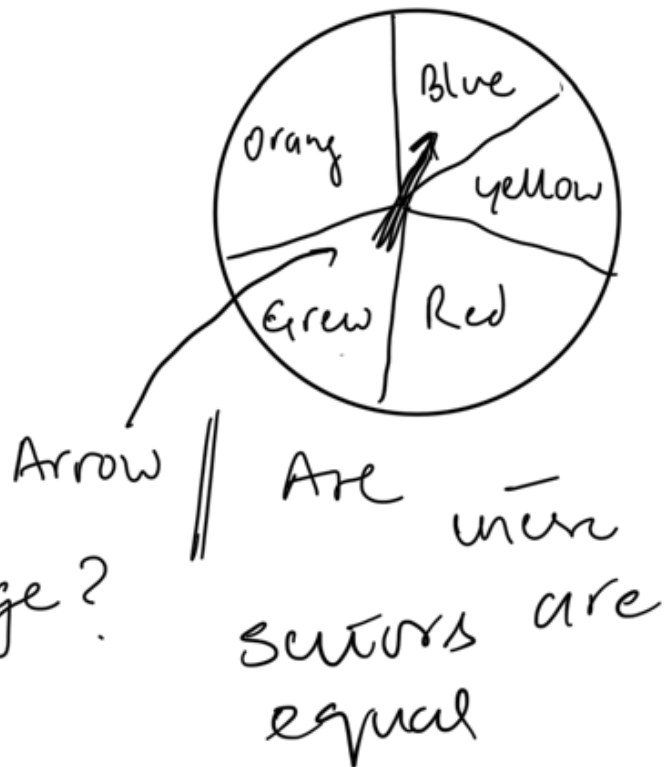
$$P(\text{Winning}) + P(\text{Losing}) = 1$$

$$0.43 + P(\text{Losing}) = 1$$

$$P(\text{Losing}) = 1 - 0.43 \\ = 0.57$$

Problem-2

What is the probability of the arrow falling on Orange?



Since all the sectors have an equal area.

the probabilities are also equal.

$$P(\text{Orange}) + P(\text{Blue}) + P(\text{Green})$$

$$+ P(\text{Red}) + P(\text{Yellow}) = 1$$

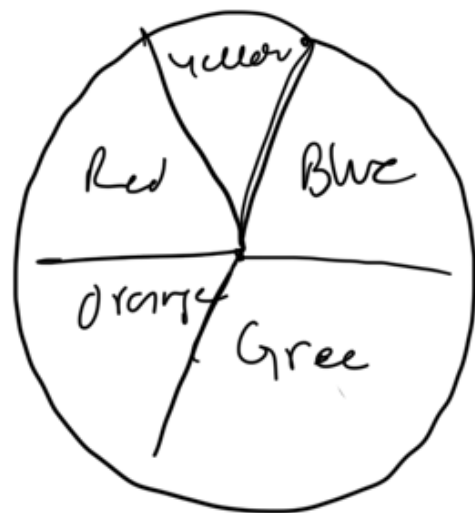
$$5 P(\text{Orange}) = 1$$

$$P(\text{Orange}) = 1/5 = 0.2.$$

Problem-3 what is the probability of 'NOT' Landing on Blue.

$$P(\text{NOT Blue})$$

$$= 1 - 0.20$$



$P(\text{Outcome does NOT Happening})$

$$= 1 - P(\text{Outcome happens}).$$

Problem-7

You have 6 types
of ice cream and 4 types
of toppings.

how many combinations
can you make?

two Events

ice cream choice [6]

topping choice [4].

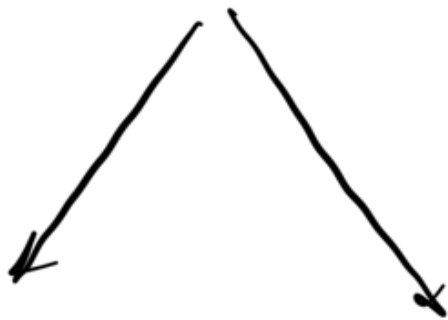
Total ways this
would have happened $\rightarrow 6 \times 4 = 24$.

Compound Event.

$\Rightarrow$ 2 win tosses [Compound
Event].

COMPOUND EVENTS

two or more events
taking place together.



Events influence
each other

(Dependent Events)

Event don't
influence
each other.

(Independent
Events.)

① two coin tosses

[they don't influence
each other].

② Picking two cards out
of a deck.

→ i ask you to pick

the 'king of hearts'
out of a deck in
two tries.

(These events influence each other).

(i) Deck of cards has 54
cards

$$P(\text{King of hearts } 1^{\text{st}}) = \frac{1}{54}$$

$$P(\text{King of hearts } 2^{\text{nd}}) = \frac{1}{53}$$

Probabilities of Independent Events.

$\Rightarrow$ The probability of two
or more independent events
taking place together is equal
to the "product" of the
individual events.

Two coin tosses

$\rightarrow$ What is the probability

of HH.

$$\begin{aligned} P(HH) &= P(H) \cdot P(H) \\ &= 0.5 \times 0.5 \\ &= 0.25 \end{aligned}$$

$$P(HTHTH) = P(H) \cdot P(T) \cdot P(H) \cdot P(T) \cdot P(H)$$

Probability of Turbo studying

& Sacha passing this class.

$P(\text{Turbo studies \& Sacha Passes})$

$$= \underline{P(\text{Turbo studies})} \times \underline{P(\text{Sacha Passes})}.$$

Probability of Sacha studying

& Sasha Passing -

$P(\text{Sasha Studies} \& \text{Sasha Passes})$

$\neq P(\text{Sasha Studies})$
 $\times P(\text{Sasha Passes}).$

$P(\text{Sasha Studies} \& \text{Sasha Passes})$

$= \underline{P(\text{Sasha Studies})} \times P(\text{Sasha Passing after studying}).$

Higher

$P(\text{Sasha Doesn't Study} \& \text{Sasha Passes})$

$= \underline{P(\text{Sasha Doesn't Study})}$
 $\times P(\text{Sasha Passes without studying}).$

lower



Problem - 1

Picking a red marble out of a bag and then picking another marble without replacing the previous one.

[Dependant].

Marble

Мрамор

Problem 2

Rolling 2, 6 Numbered die & getting a six on both of them.

[Independent].

Problem 3.

There are 5 cards numbered 1 through 5.

Sam selects a card, doesn't replace it, then selects again, what is the probability $P(1 \text{ and then } 3)$

$$P(1 \text{ and then } 3) = ?$$

[Dependent].

$$\frac{1}{5} \times \frac{1}{4} = \frac{1}{20}.$$

$$P(1 \text{ and then } 3) = P(\text{picking } 1 \text{ out of } 5) \\ \times P(\text{picking } 3 \text{ out of } 4 \text{ cards})$$

Problem - 4.

Evan has 4 red marbles,
4 green markers, 1 yellow
marker and 2 black markers
in his pencil case.

He picks one marker from the
case and doesn't replace it.

Then he picks a second marker.
What is the probability

$$(i) P(\text{black, then black}) \\ = \frac{2}{11} \times \frac{1}{10} = \frac{2}{110}$$

$$(ii) P(\text{red, then green}) \\ = \frac{4}{11} \times \frac{4}{10} = \frac{16}{110}.$$

ПЕРЕСТАНКИ

КОМБИНАЦИИ

PERMUTATIONS AND COMBINATIONS

all possible
ways to combine

all possible
ways to

ways to combine things such that the order of the combinations matters

ways to combine things such that the order doesn't matter.

PERMUTATION: A permutation is an arrangement where order 'IS' important

COMBINATION: A combination is an arrangement in which order "IS NOT" important.

Permutation with Repetition.

→ When an arrangement has a certain number of possibilities

(or outcomes) then we have
(n) choices each time.

if you have a deck of 54
cards & you have to pick a
card, you have 54 choices
in this event

Event: Picking a card out
of a deck of 54
cards

Outcomes/choices: 54.

if you have to pick two
cards with replacement, how
many choices do you have

OR

how many possible outcomes
are there in this series
of events.

...

$$54 \times 54$$

first card choice
54

3 cards with replacement

$$54 \times 54 \times 54$$

Permutation

$\Rightarrow$ No. of choice/outcome
for 1st Event X

No. of choice/outcome
for 2nd Event X

No. of choice/outcome
for 3rd Event X

$\vdots$

No. of choices/outcomes
for the nth/final Event.

Example for Permutation
without Repetition.

Let us assume that you
have 6 different coloured
buttons

you want to arrange them.

What are the possible permutations?

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720.$$

you have 720 possible permutations
or arrangements.

there is another way to write
this.

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

↗ 6 factorial

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$10! = 10 \times 9!$$

$$n! = n \times (n-1) \times (n-2) \times (n-3) \cdots 3 \times 2 \times 1$$

↗ n factorial.

So, the possible number of

arrangements for n things
without repetition is $n!$

Example for Permutation
with example

- say, you have 7
medals. You have to
put 5 of these medals
on the wall.

$$7 \times 6 \times 5 \times 4 \times 3 = \frac{7!}{2!}$$

Possible arrangements.

there is a general formula for this.

if you have " n " things

and you want to
arrange 'r' things out of
those 'n' things,
your possible arrangements
equal to

$$\frac{n!}{(n-r)!} = {}_n P^r$$

$P(n, r)$

$$n = 7$$

$$r = 5$$

$$P(7, 5) = 7P^5 = \frac{7!}{(7-5)!}$$
$$= \frac{7!}{2!}$$

$$n! \quad \text{or} \quad {}_n P^r$$

COMBINATIONS

we will pick / arrange

things without caring about
the order in which we
pick/arrange things.

Example:

you need to
make a team
of 2 people out of
3 people

People = $\{M, T, S\}$.

Let's say I use permutation

$3 \times 2 = 6$ total
Permutation

Permutation

(M, T) (M, S) (T, M) (T, S) (S, M) (S, T)



Order won't matter.
here i am only interested
in combinations,

$\left[\{M, T\}, \{M, S\}, \{S, T\} \right]$

Only 3 possible sample combinations.

Probability of SM on
the same team.

$$P(\{S, M\}) = \frac{1}{3}.$$

↑ Probability

We can formalize the
calculation for combinations

When we are picking
'r' things out of 'n' things

→ combination

$${}^nC_r = \frac{{}^nP_r}{r!} = \frac{n!}{(n-r)!r!}$$

$$n=3$$

$$r=2$$

$${}^3C_2 = \frac{3!}{(3-2)!2!} = \frac{3!}{1!2!}$$

$$= \frac{3!}{1 \times 2!} = \frac{3 \times \cancel{2} \times 1}{\cancel{2} \times 1} = 3.$$

$$0! = 1$$

0 factorial equals to 1.

Example

A team is choosing its uniform colors. They can choose from red, gold, black, green, purple, silver, blue, orange, white, red. How many different 2-colour uniforms could the team have?

Combination or order.

Sasha's hoodies are

say → Dark Blue, Light Blue, Black, Yellow.

Yellow, Black, Dark Blue, Light Blue
↑ say

$$n = 10$$

$$r = 2.$$

$${}^{10}C^2 = \frac{10!}{(10-2)! 2!}$$

$$= \frac{10!}{8! \cdot 2!} = \frac{10 \times 9 \times \cancel{8!}}{\cancel{8!} \times 2!}$$

$$= \frac{\cancel{10} \times 9}{\cancel{2} \times 1} = 5 \times 9 = \underline{\underline{45}}$$